Table 1

IT phase	Inquiries/Functions/Operations
Root cause analysis	▪ What is the scope and impact of the issue? ▪ Is the issue specific to one client, group of clients, entire site, or many locations? ▪ Is there any common pattern with respect to the failed devices? ▪ Are there any X, Y, Z events recorded in server logs? ▪ How is the storage performance while users face performance degradation? ▪ Are there any suspicious patterns in CPU, RAM, resource utilisation time series data, etc? ▪ Were there any patches applied on any component in the past 24 hours? ▪ Did new firewall configurations block critical ports? ▪ How many connection failures were there in west region sites? ▪ When X event occurred in component N, what was the database response time?
Regular IT operations	▪ Create an upgrade schedule for all regions. ▪ Where should I host the new desktop/app among the available 20 host servers? ▪ How many extra servers are needed in the next quarter? ▪ I am starting patching, informing active users, and sending reminders till they log off. ▪ Detect anomalies. ▪ With API driven interfaces, trigger external tasks like creating a helpdesk ticket, without manual intervention. ▪ Create a management report.

Typically, virtual assistants can do the following:

- On-demand interaction
- Context-driven recommendation
- Automated task execution
- Real-time reporting
- Pull the requested data quickly from multiple sources
- Remove noise and highlight contextual information
- Apply AI algorithms to find required results
- Search for anomaly patterns in huge data
- Correlate events from multiple sources to find relations and cumulative issues
- Enable automation at scale
- Run audit scripts and eliminate false negatives
- Analyse time series data for variations
- Analyse segment failures

AI based digital assistants can improve regular IT operations as well. They can:

- Estimate future capacity requirements based on current trends
- Enable automated rolling updates and roll-backs
- Automate mundane admin tasks
- Find anomalies

Architecture layers

Figure 1 shows the blueprint of a typical digital assistant.

Teams can give commands or query the virtual assistant by voice. The non-linguistic platforms available today are mature enough to understand user utterances and translate them into meaningful intents. A conversational engine can be built with advanced cognitive computing technology that allows virtual assistants to understand and carry out multi-step requests.

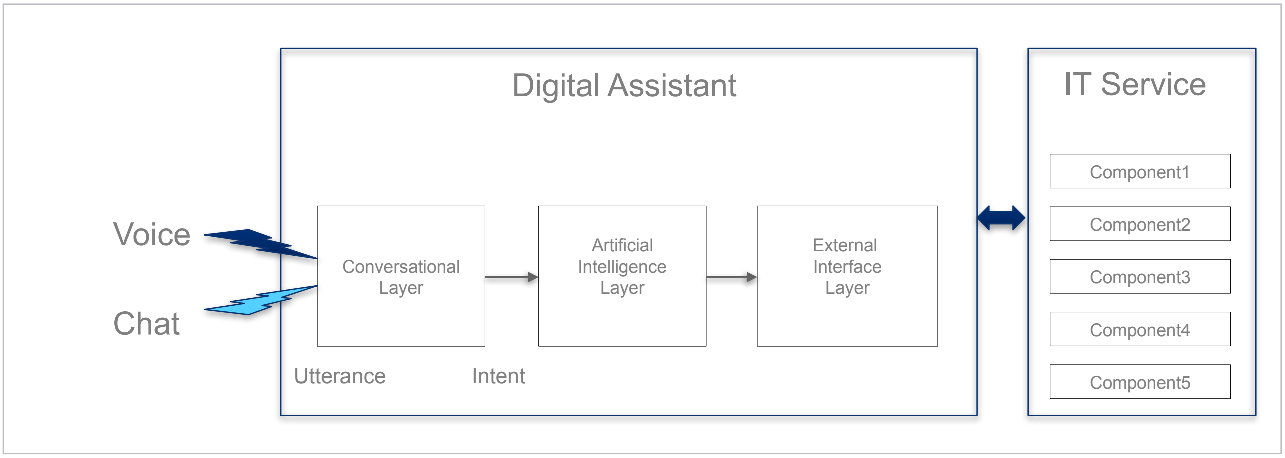

Figure 1: Digital assistant layers